Dates Lab #	LECTURE Monday, 1-3pm IRC G1	LAB MODULE ACTIVITIES T/W/Th/F 1-5pm Biological Sciences 3145	NOTES & FORMAL ASSESSMENTS
Jan 5-9	Course Orientation Research project overview	No in-person labs	
LAB 1 Jan 12-16	Lab 1 overview (Aseptic work/bacterial work & Plasmids) Reading scientific papers activity Part 1	Bacterial culture aseptic technique • Bacterial transformation • Plasmid preparation	Self-Assessment 1 **Due:** Jan 12th Lab Safety Quiz Lab 1 Quiz **Due:** Jan 12th
LAB 2 Jan 19-23	Lab 2 overview (Mammalian cell culture) Reading scientific papers activity 2	Mammalian cell culture and aseptic technique • Basic tissue culture skills • Subculturing adherent cells; counting cells • Phase microscopy Record results from lab 1	*(Last day to drop without a 'W')* Lab 2 Quiz **Due:** Jan 19th
LAB 3 Jan 26-30	Lab 3 overview (Cell stimulation SDS PAGE) Reading scientific papers activity 3	Cell stimulation, Protein Extraction and quantification; Cast resolving gel for SDS-PAGE	Lab 3 Quiz **Due:** Jan 26th
LAB 4 Feb 2-6	Lab 4 Overview (SDS-PAGE, Staining)	SDS-PAGE: • Cast stacking gel, prep protein, run gel, transfer Fluorescent labeling cultured cells **Formative feedback – BSC Aseptic technique**	Lab 4 Quiz **Due:** Feb 2nd
LAB 5 Feb 9-13	Lab 5 overview (Western blotting, Fluorescence microscopy)	Western blotting Fluorescence microscopy **Formative feedback – BSC Aseptic technique**	Lab 5 Quiz **Due:** Feb 9th Self-Assessment 2 **Due:** Feb 15th Annotated Bibliography Worksheet and Concept Map **Due:** Feb 15th
Feb 16-20	FAMILY DAY/READING WEEK		
LAB 6 Feb 23-27	Making figures and writing figure legends Data analysis workshops: Image J – Analyzing images Image J – Western blot quantification	**Practical Assessment – Tissue culture work in the BSC**	IN LAB Formal assessment – BSC Aseptic technique
LAB 7 Mar 2-6	Lab 7 overview (ELISA)	ELISA	Lab 7 Quiz **Due:** Mar 6th
LAB 8 Mar 9-13	Lab 8 Overview (RNA Extraction and qPCR) Data analysis Workshop - ELISA	RNA Extraction and qPCR	
LAB 9 Mar 16-20	Lab 9 overview (Splenic cell isolation and surface antigen labeling) Lab 10 overview (Flow cytometry) Data analysis Workshop: qPCR Data analysis and formal figure support	Splenic cell isolation and surface antigen labeling	Formal Figure and figure legend (individual) **Due:** Mar 22nd
LAB 10 Mar 23-27	Scientific Writing Storyboarding our manuscript - Recap of lab activities	Field trip to the LSI Flow Cytometry Facility Flow cytometry and Data Analysis workshop – (FlowJo)	
NO LAB Mar 30-Apr 3	Flow Cytometry debrief Flow Cytometry Assignment support	FLEX WEEK – NO LABS In lab – data analysis and writing support	
NO LAB Apr 6-10	Preparing your manuscript Peer Review Session	FLEX WEEK – NO LABS In lab – data analysis and writing support	Peer Assessment (Pair work survey) **Due:** April 10th Flow Cytometry Assignment **Due:** April 12th Manuscript (in pairs) **Due:** April 19th (date may change depending on final exam date)
Final Exam period Apr 14-25	Final exam date TBA (exam schedule released Mid-Feb)		**Due on final exam day:** Lab Notebook Self-Assessment 3